

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: HOPE | Context: US and South Asian Mental Health Counseling Practitioners (OMHC/Teletherapy)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The HOPE input ontology is explicitly scoped to dyadic, structured clinical counseling sessions (CBT, child, family) [Q22, Q26], with the taxonomy reflecting therapist-driven speaker asymmetry [Q125]. The deployment explicitly extends to peer support and crisis intervention, which involve communicative patterns the taxonomy was not designed to capture. Peer support contexts lack the formal clinical hierarchy embedded in the speaker-initiative/responsive structure, and crisis intervention involves safety-planning, de-escalation, and risk-stratification acts with no dedicated representation — these would default to IRQ, YNQ, or the residual GC catch-all [Q44]. The 988 Lifeline's structured three-phase crisis model (Explore/Clarify/Plan) does not map onto HOPE's flat 12-label taxonomy [WEB-2]. No crisis-specific dialogue-act benchmark exists to fill the gap [WEB-20]. ESConv's 8-strategy framework for peer/social support [WEB-18] is structurally different from HOPE's taxonomy, indicating that the field treats peer support as ontologically distinct. Given peer support and crisis are HIGH-priority deployment contexts, this is a substantial ontological gap. The user's elicitation acknowledges this directly (A1).

ID	Evidence
Q22	"In this section, we present our dialogue-act classification dataset, called HOPE. In total, we annotated ~ 12.9K utterances with 12 dialogue-act labels carefully designed to cater to the requirements of a counseling session." (p.3)
Q26	"The data collection process provides us 12.9K utterances from 212 counseling therapy sessions – all of them are dyadic conversations only." (p.3)
Q44	"General Chit-Chat (GC): Other utterances that do not belong to any of the above labels are tagged as GC, possibly because of the vagueness and the lack of sense in the context of the conversation." (p.4)
Q125	"This is particularly apparent as a majority portion of the counseling conversation is driven by the therapist, thus the speaker-initiative dialogue-acts have higher relevance with the therapist utterances." (p.11)
WEB-2	988 Lifeline three-phase crisis model uses communicative acts not represented in HOPE labels (https://988lifeline.org/wp-content/uploads/2023/02/FINAL_988_Suicide_and_Crisis_Lifeline_Suicide_Safety_Policy_-3.pdf)
WEB-18	ESConv 8-strategy framework for peer support is structurally distinct from HOPE (https://aclanthology.org/2021.acl-long.269/)
WEB-20	Crisis Text Line research uses discourse analysis rather than HOPE-compatible dialogue-act labels; no crisis-specific dialogue-act benchmark identified (https://www.crisistextline.org/data-philosophy/research-impact/)

Strengths

- Taxonomy is explicitly designed for counseling rather than borrowed from generic dialogue-act schemes, with hierarchical organization (initiative/responsive/general) tailored to therapy [Q27, Q29]
- Speaker-aware modeling (SPARTA) reflects the therapist-patient asymmetry that does hold within structured clinical therapy use cases in scope [Q125]
- The taxonomy is appropriate for the structured-therapy portion of the deployment (CBT/child/family in US contexts), which forms a substantial part of the practitioner workflow

ID	Evidence
Q27	"Since the counseling conversations have inherent differences with the standard conversations (such as SwitchBoard dataset conversation), it demands a carefully designed set of dialogue-act labels capable of catering the requirements of counselling conversations. Hence, we, in consultation with therapist and counselling experts, design a set of 12 dialog-act labels that are arranged in a hierarchy." (p.3)

Q29	"Each utterance in the dialogue belongs to one of the three categories – speaker initiative, speaker responsive, and general." (p.3)
Q125	"This is particularly apparent as a majority portion of the counseling conversation is driven by the therapist, thus the speaker-initiative dialogue-acts have higher relevance with the therapist utterances." (p.11)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories include structured clinical therapy acts (covered), peer support acts (mutual disclosure, emotional co-regulation, informal acknowledgment), and crisis-specific acts (safety-planning probes, lethality assessment, de-escalation, risk stratification). The deployment scope explicitly includes all three [elicitation A1].

IO-2: Yes — peer support and crisis-specific dialogue-act categories are absent. HOPE's taxonomy is built for 'patient and therapist' dyads in 'social counselling' [Q7, Q8] and for 'mental-health counseling session' [Q17], with no peer or crisis coverage. The 988 Lifeline's distinct conversational phases [WEB-2] and ESConv's separate strategy taxonomy [WEB-18] confirm that peer/crisis acts require different representational frameworks.

IO-3: No clearly irrelevant categories were identified — all 12 labels remain at least nominally applicable to general counseling dialogue. The risk is underrepresentation, not irrelevance.

IO-4: Major content-validity gaps: (a) crisis-specific safety-planning and risk-assessment acts have no dedicated label and would be subsumed into IRQ/YNQ/GC, indistinguishably from routine acts; (b) peer support emotional co-regulation likely collapses into the already-overloaded ACK or residual GC [Q44]; (c) the speaker asymmetry baked into the taxonomy [Q125] does not map onto peer dyads.

ID	Evidence
Q7	"Unlike general goal-oriented dialogues, a conversation between a patient and a therapist is considerably implicit, though the objective of the conversation is quite apparent." (p.1)
Q8	"The nature of conversations in a social counselling setting is particularly distinct as compared to a conventional chit-chat or goal-oriented conversations." (p.1)
Q17	"We propose twelve dialogue-act labels that are aligned with mental-health counseling session." (p.2)
Q22	"In this section, we present our dialogue-act classification dataset, called HOPE. In total, we annotated ~ 12.9K utterances with 12 dialogue-act labels carefully designed to cater to the requirements of a counseling session." (p.3)
Q26	"The data collection process provides us 12.9K utterances from 212 counseling therapy sessions – all of them are dyadic conversations only." (p.3)
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Information Gaps

- No empirical validation of how HOPE labels behave on peer support or crisis-register text
- No ontology defined for crisis-specific acts such as safety-planning probes or lethality assessment

Requires Expert Verification

- Whether peer support practitioners would recognize their communicative patterns in HOPE's taxonomy
- Whether crisis intervention staff would consider the IRQ/YNQ collapse acceptable for their workflow

Remediation

Gap 1: No dedicated label coverage for crisis-specific (safety-planning, risk-stratification, de-escalation) or peer-support (mutual disclosure, emotional co-regulation) communicative acts; these collapse into IRQ/YNQ/GC.

Recommendation: Before deploying in crisis or peer-support contexts, extend the taxonomy with crisis-specific labels informed by the 988 Lifeline three-phase model [WEB-2] and peer-support strategy labels informed by ESConv [WEB-18]. Validate the extended taxonomy on a held-out sample of crisis and peer-support transcripts before integrating into production feedback workflows.

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WEB-2	988 Lifeline three-phase crisis model uses communicative acts not represented in HOPE labels (https://988lifeline.org/wp-content/uploads/2023/02/FINAL_988_Suicide_and_Crisis_Lifeline_Suicide_Safety_Policy_-3.pdf)
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